Equations of circles

1 **Graph B**

2

a No solution provided

b Quadrilateral

c Circle

3

a
- i $(0, 0)$
- ii 2 units

b
- i $(3, 2)$
- ii 3 units

c
- i $(1, -2)$
- ii 8 units

d
- i $(-3, -1)$
- ii 5 units

4 Every point on the circle is exactly 5 units away from the point $(0, 0)$.

5

a 1

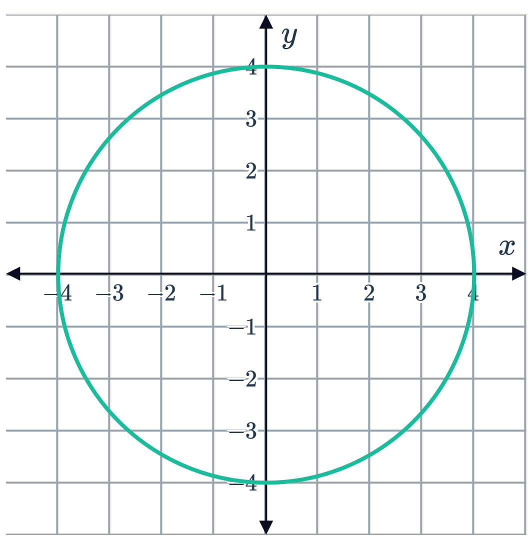
b 0

6 $x^2 + y^2 = 49$

7

a

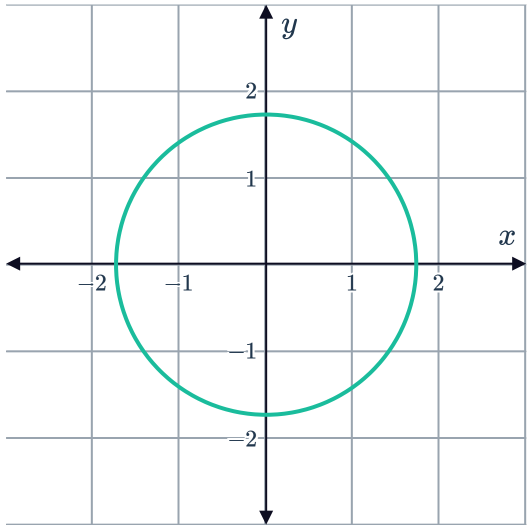
i



ii $x^2 + y^2 = 16$

b

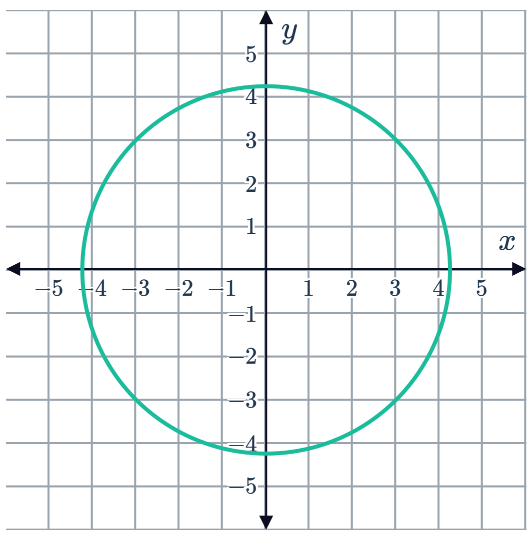
i



ii $x^2 + y^2 = 3$

c

i



ii $x^2 + y^2 = 18$

8

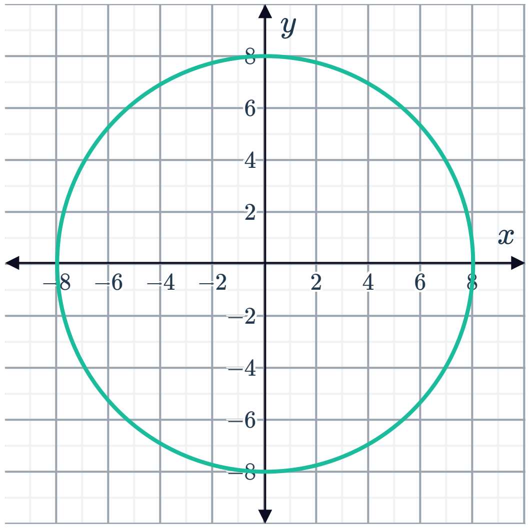
a $(0, 0)$

b 8 units

c $x = 8, -8$

d $y = 8, -8$

e



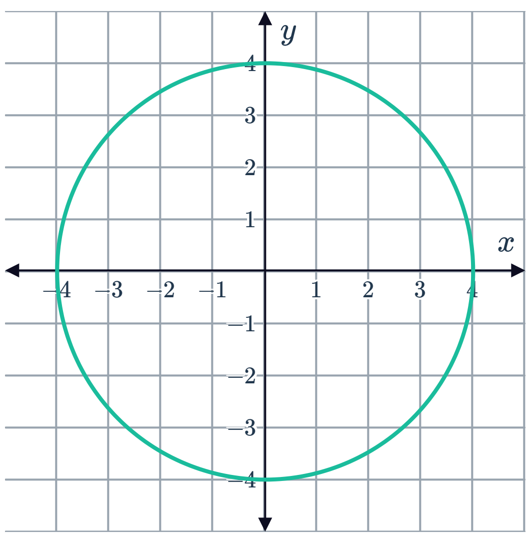
9 $4\sqrt{2}$ units

10

a

i 4 units

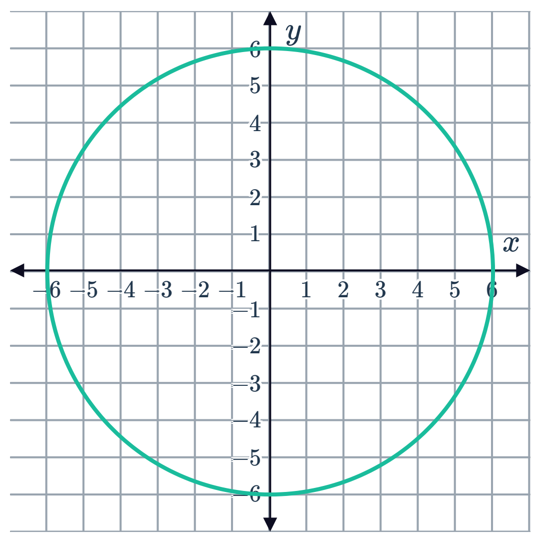
ii



b

i 6 units

ii



11 $x^2 + y^2 = 4$

12

a $5\sqrt{5}$ units

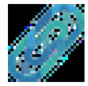
b $x^2 + y^2 = 125$

13

a $x^2 + y^2 = 36$

b 36π units^2

14 **a** 14 units



$y = \pm\sqrt{48}$

15 Inside

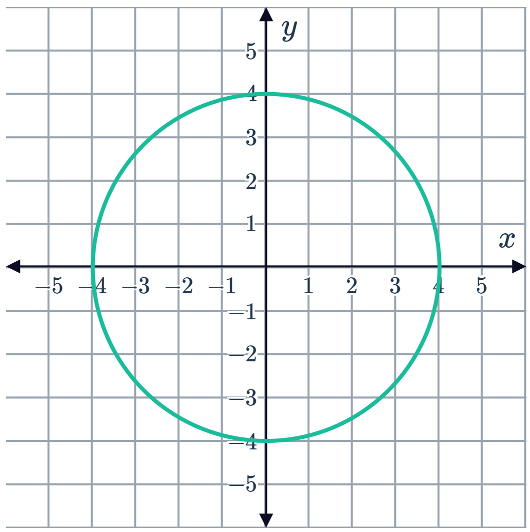
16

a

i $(0, 0)$

ii 4 units

iii

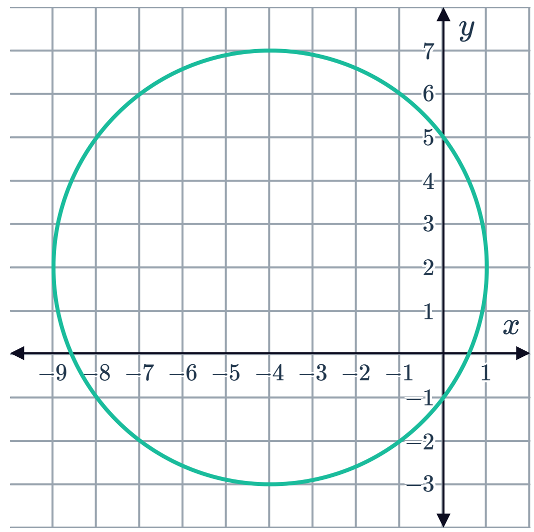


b

i $(-4, 2)$

ii 5 units

iii

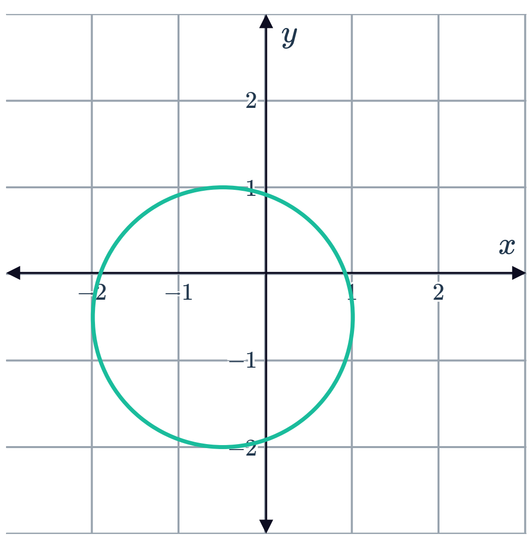


c

i $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

ii $\frac{3}{2}$ units

iii

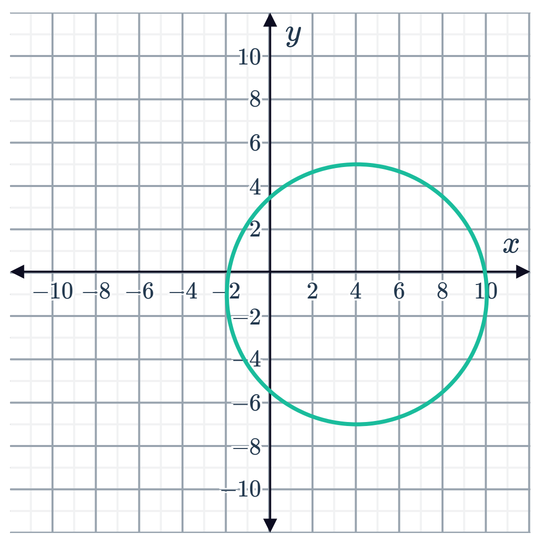


d

i $(4, -1)$

ii 6 units

iii

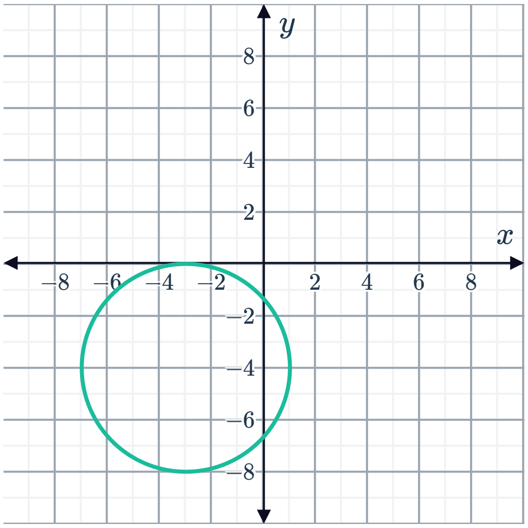


e

i $(-3,-4)$

ii 4 units

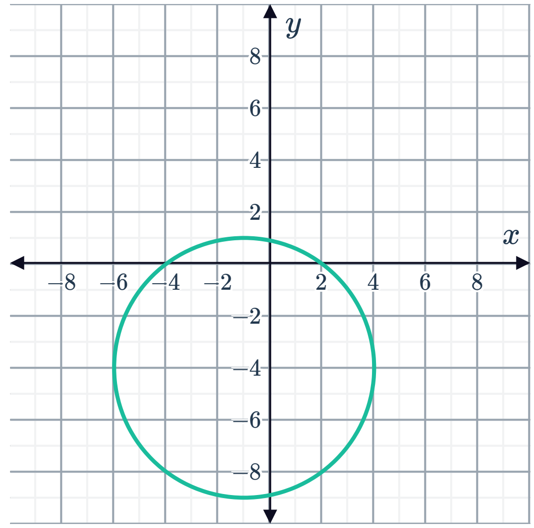
iii



i $(-1,-4)$

ii 5 units

iii

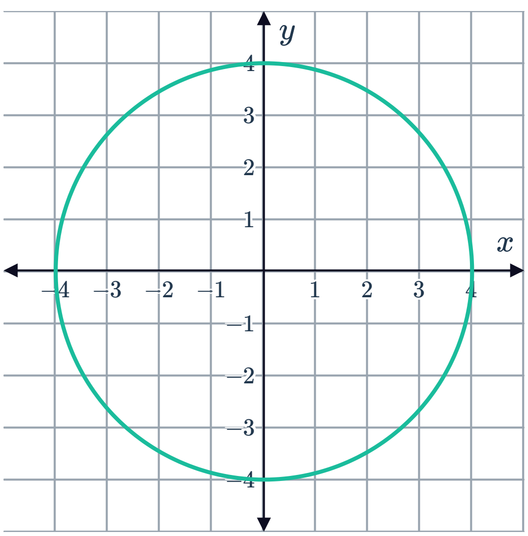


g

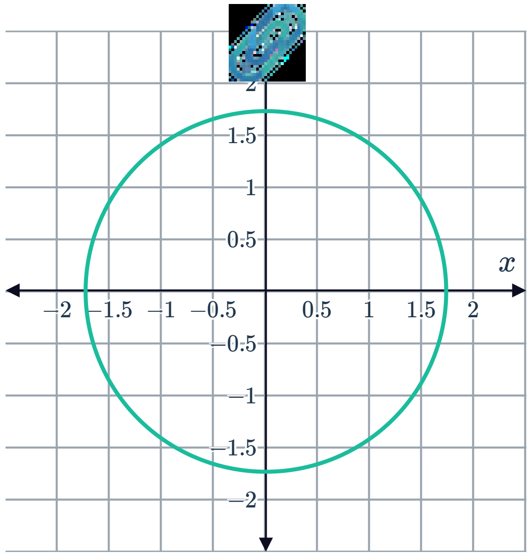
i $(0,0)$

ii 4 units

iii



17



18 $\sqrt{353}$ units

19

a

i $\left(-3.1, \frac{3}{2}\right)$ ii $\frac{3}{4}$ units

b

i $(6, 6)$ ii $2\sqrt{3}$ units

c

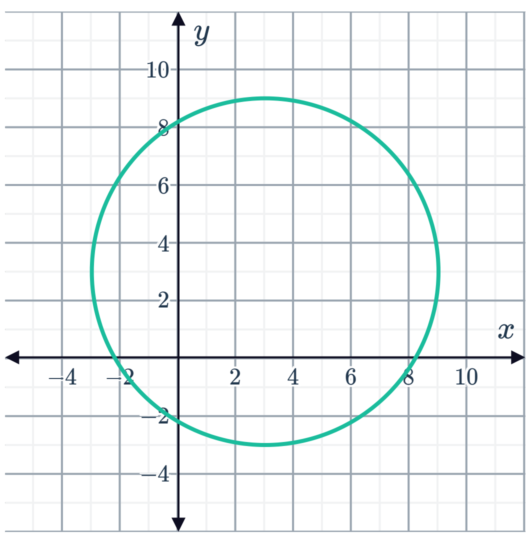
i $(0.7, -4.4)$ ii $\sqrt{3}$ units

d

i $(8, -1)$ ii $\sqrt{13}$ units

20

a



b $(x-3)^2 + (y-3)^2 = 36$

21

a

i $(-3,-3)$

ii 3 units

iii $(x+3)^2+(y+3)^2=9$

b

i $(1,3)$

ii 6 units

iii $(x-1)^2+(y-3)^2=36$

22

a $(x-1)^2+(y+3)^2=25$

b $(x+5.7)^2+y^2=9$

23 Two

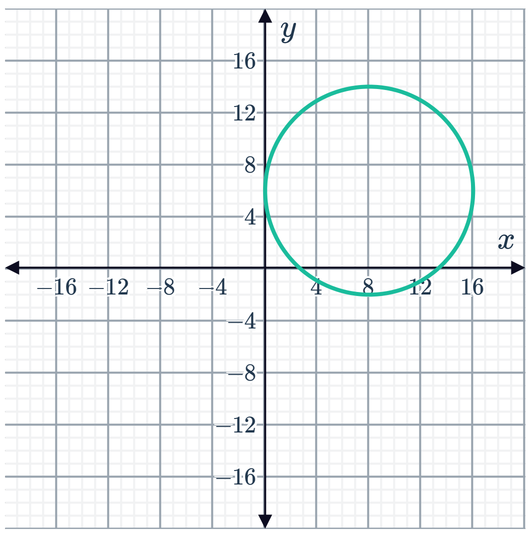
24

a $(x-8)^2+(y-6)^2=64$

b No

25

a



b $x^2-4x+y^2+12y+31=0$

26

a Outside

b On

c Inside

27

a Inside

b On

c On

Additional questions

29

a

i $(0, 0)$

ii 5 units

iii 10 units

iv $x^2 + y^2 = 25$

b

i $(1, -2)$

ii 4 units

iii 8 units

iv $(x-1)^2 + (y+2)^2 = 16$

30

a $(2, -3)$

b $r = 5\sqrt{3}$ units

c $r = 10\sqrt{3}$ units

31

a $x^2 + (y-6)^2 = 9$

b $\left(x - \frac{17}{2}\right)^2 + y^2 = 15$

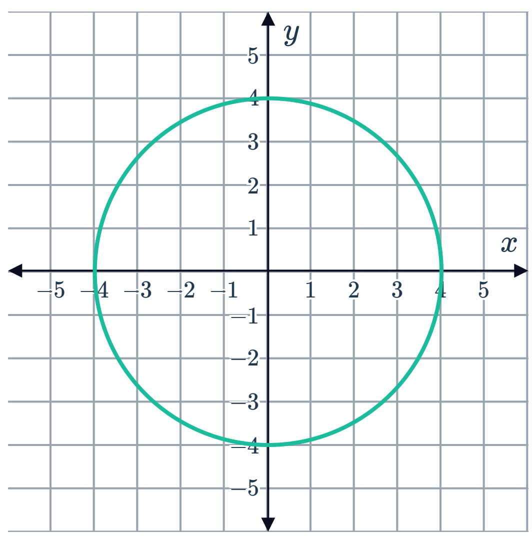
32

a

i $(0, 0)$

ii 4 units

iii



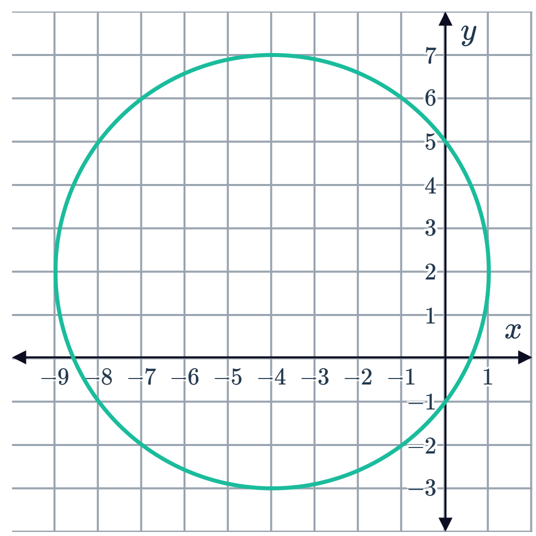
iv Domain $= [-4, 4]$
Range $= [-4, 4]$

b

i $(-4, 2)$

ii 5 units

iii



iv Domain $= [-9, 1]$
Range $= [-3, 7]$

33

a 13 units

b $(x+2)^2 + (y-5)^2 = 169$

34

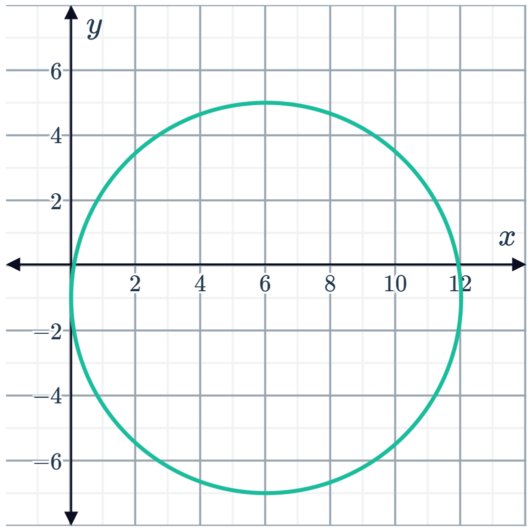
a Inside **b** On **c** Outside **d** On

35

a

i $(x-6)^2+(y+1)^2=36$

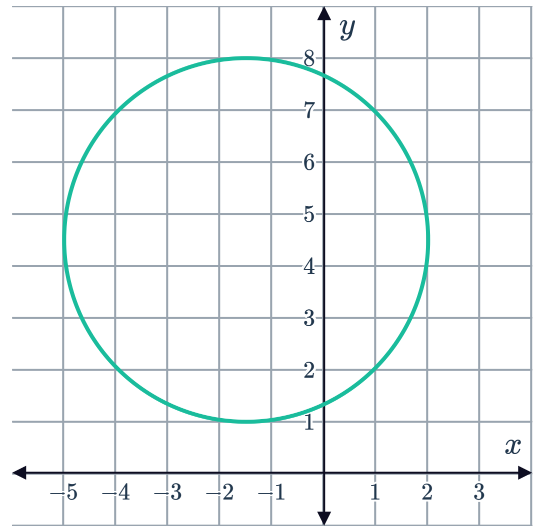
ii



b

i $\left(x+\frac{3}{2}\right)^2+\left(y-\frac{9}{2}\right)^2=\frac{49}{4}$

ii



36 $(x-25)^2+(y-30)^2=400$

37

a $(49.5, 37.5)$

b 9.15 m

c $(x-49.5)^2+(y-37.5)^2=83.7225$

d Domain $=[40.35, 58.65]$
Range $=[28.35, 46.65]$

38

a Domain: $[5, 19]$, Range: $[1, 15]$

b Domain: $[14, 26]$, Range: $[2, 14]$

c 5 cm

d 85.19 cm^2

39

a $(x-4)^2+(y+3)^2=41$

b Given:

- Equation of circle: $(x-4)^2+(y+3)^2=41$
- Point: $P\,(8,3)$

To prove: Distance between circle center and $P\,(8,3)$ is greater than $\sqrt{41}$

1. Find the center of the circle

$$(x-4)^2+(y+3)^2=41 \qquad \text{Given}$$

$$C=(4,-3) \qquad \text{Equation of the circle}$$

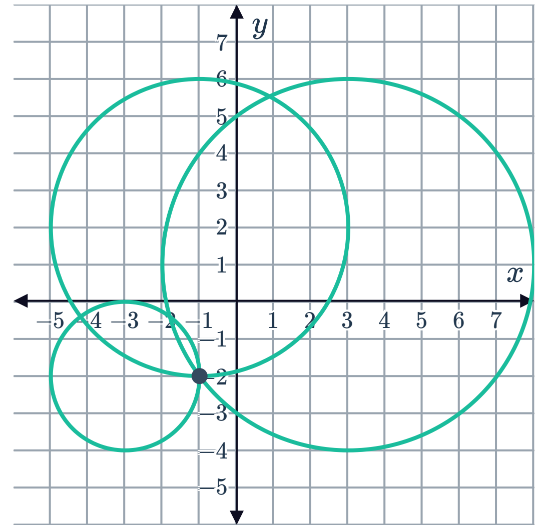
2. Find distance between center and $C\,(8,3)$

$$
\begin{aligned}
CP &= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} && \text{Distance formula}\\
&= \sqrt{(8-4)^2+(3--3)^2} && \text{Substitution}\\
&= \sqrt{(4)^2+(6)^2} && \text{Simplifying}\\
&= \sqrt{16+36} && \text{Simplifying}\\
&= \sqrt{52} && \text{Simplifying}
\end{aligned}
$$

3. Show distance between circle center and $P\,(8,3)$ is greater than $\sqrt{41}$
   $CP=\sqrt{52}>\sqrt{41}$
   Therefore, P is outside the circle.

40

a As the point of intersection of the three circles centered at the receiving stations with radii as the distances, the epicenter is at $(-1, -2)$.



b The point $(8, -7)$ is outside the circle with center $(-1, -2)$ with radius 10, so it would not feel the earthquake.